

(assuming the integrals exist). A MC estimate can be obtained as follows:

1. generate m pseudo-random realizations, $\mathbf{x}_{0:t}^i$ from $p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})$, $i = 1, \dots, m$
2. evaluate f for each realization and compute the arithmetic average of the results, $\hat{E}(f(\mathbf{X}_{0:t})|\mathbf{y}_{1:t}) = (1/m) \sum_{i=1}^m f(\mathbf{x}_{0:t}^i)$.

Under independent sampling this average converges (almost surely) to $E(f(\mathbf{X}_{0:t})|\mathbf{y}_{1:t})$ as m approaches infinity. This also holds if realizations are stationary and ergodic though not necessarily independent. We note that the rate of convergence is independent of the dimensionality of the integrand (for details, see [33]). We can also approximate the distribution $p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})$ by:

$$\hat{p}(\mathbf{x}_{0:t}|\mathbf{y}_{1:t}) \equiv p^m(\mathbf{x}_{0:t}|\mathbf{y}_{1:t}) = \frac{1}{m} \sum_{i=1}^m \delta_{\mathbf{x}_{0:t}^i}$$

where $\delta_{\mathbf{x}_{0:t}^i}$ is a Dirac-delta mass on the trajectory $\mathbf{x}_{0:t}^i$. In practice, one typically considers kernel density estimates of the posterior (e.g., [36,41]).

4.2. Importance sampling Monte Carlo

When direct simulation from $p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})$ is very difficult, *Importance Sampling Monte Carlo (ISMC)* is suggested. The idea is that we consider another distribution, say $g(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})$, that is comparatively easy to sample. We generate m samples $\mathbf{x}_{0:t}^i$ from $g(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})$ and evaluate f for each. A standard procedure for estimating $E(f(\mathbf{X}_{0:t})|\mathbf{y}_{1:t})$ is to weight each sample or ensemble member to adjust for the fact that they are not from the target posterior as follows:

$$\hat{E}(f(\mathbf{X}_{0:t})|\mathbf{y}_{1:t}) = \sum_{i=1}^m w^i f(\mathbf{x}_{0:t}^i),$$

where the weights w^i are defined by

$$w^i = \frac{p(\mathbf{x}_{0:t}^i|\mathbf{y}_{1:t})/g(\mathbf{x}_{0:t}^i|\mathbf{y}_{1:t})}{\sum_{j=1}^m p(\mathbf{x}_{0:t}^j|\mathbf{y}_{1:t})/g(\mathbf{x}_{0:t}^j|\mathbf{y}_{1:t})}. \quad (33)$$

In this case, we can approximate the posterior distribution by

$$p^m(\mathbf{x}_{0:t}|\mathbf{y}_{1:t}) = \sum_{i=1}^m w^i \delta_{\mathbf{x}_{0:t}^i},$$

or, in practice, by a kernel-density approximation (e.g., [41]).

While efficient selection of the importance function g is the subject of much research, we only consider the following common approach. Assume that g is the prior or forward model distribution:

$$g(\mathbf{x}_{0:t}|\mathbf{y}_{1:t}) = p(\mathbf{x}_{0:t}).$$

Thus, given a sample from $p(\mathbf{x}_0)$, one obtains MC trajectories by simulating from the forward model. Applying (33) for this

g , we obtain

$$w^i \propto \frac{p(\mathbf{x}_{0:t}^i|\mathbf{y}_{1:t})}{p(\mathbf{x}_{0:t}^i)} \propto p(\mathbf{y}_{1:t}|\mathbf{x}_{0:t}^i). \quad (34)$$

Such an ensemble-based approach for obtaining samples from the posterior is sometimes referred to as an *ensemble smoother*. The crucial point to note here is that this approach can be implemented even if we cannot find the normalizing constant of the target posterior $p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})$.

4.3. Sequential Monte Carlo

In principle, a general sequential MC algorithm follows from the usual sequential update distributions (13) and (14).

As mentioned in the previous section, the choice of the prior distribution as the importance distribution gives weights proportional to the likelihood. We note that these weights can be computed iteratively as follows. Let the weight on ensemble member i at time t be denoted w_t^i . Since the data are assumed to be conditionally independent across time, it follows that

$$w_t^i \propto p(\mathbf{y}_{1:t}|\mathbf{x}_{0:t}^i) \propto p(\mathbf{y}_t|\mathbf{x}_t^i) w_{t-1}^i. \quad (35)$$

Assume we have m samples (or “particles”) from the analysis (posterior) distribution at time $t-1$, $p(\mathbf{x}_{t-1}|\mathbf{y}_{1:t-1})$, denoted by $\mathbf{x}_{t-1|t-1}^i$, $i = 1, \dots, m$. The analysis distribution at time $t-1$ can then be approximated by:

$$p^m(\mathbf{x}_{t-1}|\mathbf{y}_{1:t-1}) = \sum_{i=1}^m \delta_{\mathbf{x}_{t-1|t-1}^i} w_{t-1}^i,$$

where w_{t-1}^i are the normalized ISMC weights for time $t-1$. As with basic Monte Carlo, one often considers a kernel-density estimate of this distribution in practice. This, can then be used to estimate the forecast distribution as the following mixture:

$$p^m(\mathbf{x}_t|\mathbf{y}_{1:t-1}) = \sum_{i=1}^m p(\mathbf{x}_t|\mathbf{x}_{t-1|t-1}^i) w_{t-1}^i.$$

Using the fact that the weights can be updated as in (35), the analysis distribution at time t is then given by:

$$\begin{aligned} p^m(\mathbf{x}_t|\mathbf{y}_{1:t}) &\propto p(\mathbf{y}_t|\mathbf{x}_t) \sum_{i=1}^m p(\mathbf{x}_t|\mathbf{x}_{t-1|t-1}^i) w_{t-1}^i \\ &= \sum_{i=1}^m p(\mathbf{x}_t|\mathbf{x}_{t-1|t-1}^i) w_t^i. \end{aligned}$$

Unfortunately, when the state process or data vectors are of relatively high dimension, the importance weights will degenerate as t increases (i.e., only a few, or one, sample gets all of the weight) and the posterior distribution is not adequately represented by the sample. Various practical solutions to this problem have been considered. We briefly (and generally) discuss a few of these in the following subsections.

4.3.1. Particle filtering

A comprehensive overview of the particle filtering literature can be found in [13]. As mentioned above, overcoming the